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CIS 343
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Lab1: Scanning

Apologies for the long report, I have lots of test cases.

This report includes details on Hashbrown, the programming language I am creating this semester.

1. Regular expressions and design choices.

Hashbrown follows the following regular expressions:

- Number literals: `[0-9]+(\.[0-9]+)?([eE][+-]?[0-9]+)?`
- String literals: `"[^"]*"`
- Identifiers: `[A-Za-z_][A-Za-z0-9_]*`

I chose to use `&&` and `||` instead of the words 'and' and 'or', because I am a big fan of C-style syntax. On the other hand, I love Python's `#` comments, so I chose to implement those, and I also chose to use `/#` and `*/` for multiline comments. I have unique error messages when the scanner encounters a lone `&` or `|` character: it says "did you mean '&&'?" instead of simply saying unexpected character.

2. Setup. To run my interpreter, run the following commands

```
cmake -B build
cmake --build build
```

one after another in the root directory of my project. This compiles every file and produces the executable at

```
build/hashbrown
```

You can then run it by typing

```
./build/hashbrown
```

3. Test Cases. There are 16 test cases for the normal interpreter mode where it is fed a file, and 8 test cases for the REPL mode.

The test cases are named according to what they test for. I have printed them below, with the expected output and the source code one after another for each.

```
test/lab1/cases/01_all_tokens.expected
```

```
LEFT_PAREN ( null
RIGHT_PAREN ) null
LEFT_BRACE { null
RIGHT_BRACE } null
COMMA , null
DOT . null
PLUS + null
MINUS - null
STAR * null
SLASH / null
SEMI_COLON ; null
AMPERSANDS && null
VERTS || null
REMAINDER % null
BANG ! null
BANG_EQUAL != null
EQUAL = null
EQUAL_EQUAL == null
LESS < null
LESS_EQUAL <= null
GREATER > null
GREATER_EQUAL >= null
NUMBER 123 123.0
STRING "hello" hello
IDENTIFIER identifier null
CLASS class null
ELSE else null
FALSE false null
FOR for null
IF if null
NIL nil null
RETURN return null
TRUE true null
WHILE while null
EOF null
test/lab1/cases/01_all_tokens.txt
( ) { } , . + - * / ; && || %
! != = == < <= > >=
123
"hello"
identifier
```

```
class else false for if nil return true while
test/lab1/cases/02_operators.expected
BANG ! null
BANG_EQUAL != null
EQUAL = null
EQUAL_EQUAL == null
LESS < null
LESS_EQUAL <= null
GREATER > null
GREATER_EQUAL >= null
PLUS + null
MINUS - null
STAR * null
SLASH / null
EOF null
test/lab1/cases/02_operators.txt
! !=
= ==
< <=
> >=
+ - * /
test/lab1/cases/03_punctuation.expected
LEFT_PAREN ( null
RIGHT_PAREN ) null
LEFT_BRACE { null
RIGHT_BRACE } null
COMMA , null
DOT . null
SEMI_COLON ; null
EOF null
test/lab1/cases/03_punctuation.txt
( ) { } , . ;
test/lab1/cases/04_keywords.expected
CLASS class null
ELSE else null
FALSE false null
FOR for null
IF if null
NIL nil null
RETURN return null
TRUE true null
WHILE while null
EOF null
test/lab1/cases/04_keywords.txt
class
else
false
```

```
for
if
nil
return
true
while
test/lab1/cases/05_almost_keywords.expected
IDENTIFIER andy null
IDENTIFIER classy null
IDENTIFIER elsewhere null
IDENTIFIER falsey null
IDENTIFIER forest null
IDENTIFIER iffy null
IDENTIFIER nilpotent null
IDENTIFIER orange null
IDENTIFIER returning null
IDENTIFIER truer null
IDENTIFIER whiley null
EOF null
test/lab1/cases/05_almost_keywords.txt
andy
classy
elsewhere
falsey
forest
iffy
nilpotent
orange
returning
truer
whiley
test/lab1/cases/06_identifiers.expected
IDENTIFIER foo null
IDENTIFIER bar null
IDENTIFIER _ null
IDENTIFIER _foo null
IDENTIFIER foo_ null
IDENTIFIER foo123 null
IDENTIFIER abc123xyz null
IDENTIFIER ABC null
IDENTIFIER ABC123 null
IDENTIFIER snake_case null
IDENTIFIER CamelCase null
IDENTIFIER __hello__ null
EOF null
test/lab1/cases/06_identifiers.txt
foo
```

```
bar
-
_foo
foo_
foo123
abc123xyz
ABC
ABC123
snake_case
CamelCase
__hello__
test/lab1/cases/07_numbers.expected
NUMBER 0 0.0
NUMBER 1 1.0
NUMBER 10 10.0
NUMBER 42 42.0
NUMBER 123456789 123456789.0
NUMBER 0.0 0.0
NUMBER 1.0 1.0
NUMBER 3.14 3.14
NUMBER 123.456 123.456
NUMBER 999.999 999.999
NUMBER 1e14 100000000000000.0
EOF null
test/lab1/cases/07_numbers.txt
0
1
10
42
123456789
0.0
1.0
3.14
123.456
999.999
1e14
test/lab1/cases/08_number_edge_cases.expected
NUMBER 123 123.0
DOT . null
DOT . null
NUMBER 123 123.0
NUMBER 1.2 1.2
DOT . null
NUMBER 3 3.0
NUMBER 1 1.0
DOT . null
DOT . null
```

```
NUMBER 2 2.0
NUMBER 0 0.0
DOT . null
NUMBER 0.0 0.0
NUMBER 1.00 1.0
EOF null
test/lab1/cases/08_number_edge_cases.txt
123.
.123
1.2.3
1..2
0.
0.0
1.00
test/lab1/cases/09_strings.expected
STRING "hello" hello
STRING "hello world" hello world
STRING ""
STRING "123" 123
STRING "true" true
STRING "hello123" hello123
STRING "hello, world!" hello, world!
STRING "special characters: ! @ # $ %" special characters: ! @ # $ %
EOF null
test/lab1/cases/09_strings.txt
"hello"
"hello world"
""
"123"
"true"
"hello123"
"hello, world!"
"special characters: ! @ # $ %"
test/lab1/cases/10_multiline_strings.expected
STRING "hello
world" hello
world
EOF null
test/lab1/cases/10_multiline_strings.txt
"hello
world"
test/lab1/cases/11_comments.expected
NUMBER 123 123.0
NUMBER 456 456.0
EOF null
test/lab1/cases/11_comments.txt
# this entire line should disappear
```

```
123 # this part is ignored
456
test/lab1/cases/12_block_comments.expected
NUMBER 123 123.0
NUMBER 456 456.0
EOF null
test/lab1/cases/12_block_comments.txt
123 /* comment */ 456
test/lab1/cases/13_whitespace.expected
NUMBER 123 123.0
NUMBER 456 456.0
NUMBER 123 123.0
NUMBER 456 456.0
NUMBER 123 123.0
NUMBER 456 456.0
NUMBER 123 123.0
NUMBER 456 456.0
EOF null
test/lab1/cases/13_whitespace.txt
123      456
123      456
123
456
123 456
test/lab1/cases/14_combined.expected
CLASS class null
IDENTIFIER Foo null
LEFT_BRACE { null
RETURN return null
TRUE true null
SEMI_COLON ; null
IF if null
LEFT_PAREN ( null
IDENTIFIER x null
GREATER_EQUAL >= null
NUMBER 10 10.0
RIGHT_PAREN ) null
LEFT_BRACE { null
IDENTIFIER x null
EQUAL = null
IDENTIFIER x null
PLUS + null
NUMBER 1 1.0
SEMI_COLON ; null
RIGHT_BRACE } null
ELSE else null
```

```

LEFT_BRACE { null
IDENTIFIER x null
EQUAL = null
IDENTIFIER x null
MINUS - null
NUMBER 1 1.0
SEMI_COLON ; null
RIGHT_BRACE } null
RETURN return null
STRING "done" done
SEMI_COLON ; null
RIGHT_BRACE } null
EOF null
test/lab1/cases/14_combined.txt
class Foo {
    return true;

    if (x >= 10) {
        x = x + 1;
    } else {
        x = x - 1;
    }

    # comment
    return "done";
}
test/lab1/cases/15_unexpected_characters.expected
[line 1] Error: Unexpected Character.
[line 2] Error: Unexpected Character.
[line 3] Error: Unexpected '&'. Did you mean '&&'?
[line 4] Error: Unexpected Character.
[line 5] Error: Unexpected Character.
[line 6] Error: Unexpected '|'. Did you mean '||'?
[line 7] Error: Unexpected Character.
[line 8] Error: Unexpected Character.
[line 9] Error: Unexpected Character.
EOF null
test/lab1/cases/15_unexpected_characters.txt
@
$
&
[
]
|
?
~
^

```

```
test/lab1/cases/16_terminated_strings.expected
[line 2] Error: Unterminated string.
EOF null
test/lab1/cases/16_terminated_strings.txt
"hello
```

```
test/lab1/repl/comments.expected
EOF null
IDENTIFIER x null
EOF null
IDENTIFIER y null
EOF null
[line 1] Error: Unterminated multiline comment.
EOF null
IDENTIFIER z null
EOF null
test/lab1/repl/comments.in
# just a comment
x # trailing comment
/# block #/ y
/# never closed
z
test/lab1/repl/empty_lines.expected
EOF null
IDENTIFIER x null
EOF null
EOF null
test/lab1/repl/empty_lines.in

x

test/lab1/repl/multiple_errors.expected
[line 1] Error: Unexpected Character.
EOF null
[line 1] Error: Unexpected Character.
EOF null
NUMBER 1 1.0
PLUS + null
NUMBER 2 2.0
EOF null
test/lab1/repl/multiple_errors.in
@
$
1 + 2
test/lab1/repl/numbers.expected
NUMBER 1e5 100000.0
```

```
EOF null
NUMBER 1.5E-3 0.0015
EOF null
[line 1] Error: Malformed exponent in number.
EOF null
IDENTIFIER x null
EOF null
test/lab1/repl/numbers.in
1e5
1.5E-3
1e
x
test/lab1/repl/operators.expected
NUMBER 10 10.0
REMAINDER % null
NUMBER 3 3.0
EOF null
IDENTIFIER a null
AMPERSANDS && null
IDENTIFIER b null
VERTS || null
IDENTIFIER c null
EOF null
[line 1] Error: Unexpected '&'. Did you mean '&&'?
IDENTIFIER a null
IDENTIFIER b null
EOF null
[line 1] Error: Unexpected '|'. Did you mean '||'?
IDENTIFIER a null
IDENTIFIER b null
EOF null
test/lab1/repl/operators.in
10 % 3
a && b || c
a & b
a | b
test/lab1/repl/recovery.expected
IDENTIFIER x null
EQUAL = null
NUMBER 1 1.0
EOF null
[line 1] Error: Unexpected Character.
EOF null
IDENTIFIER y null
EQUAL = null
NUMBER 2 2.0
EOF null
```

```
test/lab1/repl/recovery.in
x = 1
@
y = 2
test/lab1/repl/unterminated_string.expected
[line 1] Error: Unterminated string.
EOF null
IDENTIFIER x null
EOF null
test/lab1/repl/unterminated_string.in
"abc
x
```

My code passes all test cases. The script to run the test cases is at test/run_tests.sh

Extra credit:

I implemented scientific number literals and some enhanced error diagnostics.